Split Complex Lorentzian Dynamics

Part XXIX: Defect-Free Descent, Simultaneous Kernel Trivialization, and the Last Compatibility Criterion

From repeated-index rigidity to compatible whole-overlap transitions, with the global total-space construction isolated conditionally for Part XXX

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Abstract

The previous stages of the reconstruction progressively removed the major local ambiguities surrounding the split-complex Lorentz starting structure. The internal spatial carrier was closed, its double-valued group geometry was identified, ordinary and lifted frame torsors were reconstructed, a varying metric-oriented rank-three family was introduced, the correct atlas-generated topology was built, and continuous ordinary transition maps were obtained. Part XXVIII then confronted these ordinary transitions with the already certified two-to-one internal group projection. Every ordinary transition was proved locally continuously representable after refinement; two representatives differed uniquely by a continuous kernel-valued function; and on triple overlaps the internal multiplication law closed only up to a continuous locally constant central factor

$$\delta_{ijk} = (u_{jk}u_{ij})u_{ik}^{-1}.$$

For the certified projection the kernel is exactly $\{\pm 1\}$. The defect transformed under a change $u'_{ij} = \varepsilon_{ij}u_{ij}$ by

$$\delta'_{ijk} = (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk},$$

and associativity forced a quadruple-overlap consistency relation. What remained unclear was whether this triple defect was really the whole compatibility problem. In particular, the refined representatives of one fixed ordinary transition might still have carried an independent same-pair descent defect. If so, simultaneous removal of δ would not yet produce a genuine transition system on the original overlaps.

Task 29 resolves this hardening question completely. It first defines `TripleDefectFree` using the inherited Task-28 predicate over *all* index triples, including repeated indices. Those repeated instances are proved load-bearing. From $\delta = 1$ one derives identity normalization $u_{ii} = 1$, inverse normalization $u_{ji} = u_{ij}^{-1}$, and the exact internal triple law $u_{jk}u_{ij} = u_{ik}$. The potential same-pair discrepancy

$$\kappa_{ij}^{\alpha\beta} = u_{ij}^{\beta}(u_{ij}^{\alpha})^{-1}$$

is then reconstructed explicitly, proved continuous, kernel-valued and locally constant, and shown to satisfy its own identity, inverse and composition laws. The decisive theorem uses the repeated triple (i,i,j) to prove that full triple-defect freeness forces $\kappa = 1$. Thus κ is not an independent second obstruction. A permanently preserved counterexample proves that this conclusion would be false if defect-freeness were restricted to pairwise distinct indices only. The repeated-index quantification is therefore mathematical content, not formal redundancy.

Once $\kappa = 1$, the refined representatives of each ordinary transition agree on mutual patch intersections and glue uniquely to a continuous internal map on the original overlap. Task 29 then formalizes admissible continuous kernel adjustments and proves that an adjusted system is defect-free exactly when one adjustment ε solves the explicit system

$$(\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk} = 1, \qquad \text{equivalently} \qquad \varepsilon_{ik} = \varepsilon_{jk}\varepsilon_{ij}\delta_{ijk}.$$

The same adjustment automatically solves the pairwise descent problem because defect-freeness already implies $\kappa = 1$. There is no second independent compatibility equation.

The global solvability verdict is then hardened in three further directions. First, simultaneous trivializability is invariant under the arbitrary initial choice of local internal representatives. Second, it is invariant under refinement in both directions; "trivializable after refinement" is proved equivalent to trivializability of the original candidate. Third, the verdict is independent of the candidate itself once the ordinary transition system and internal projection are fixed. These results culminate in the principal theorem of Part XXIX: simultaneous trivializability of the internally derived kernel equations is equivalent to the existence of a normalized system of continuous internal transition maps on the *original* pair overlaps, projecting to the ordinary transitions and satisfying exact identity, inverse and triple-overlap laws.

The formal run contains eleven new Lean files and 2217 new Lean lines. The focused build is green at 8257 jobs; the whole project is green at 8328 jobs with zero errors. The Task-29 layer has zero warnings; all 114 whole-project warnings are inherited from untouched earlier files. Fifty-nine principal endpoints report exactly the standard baseline `propext`, `Classical.choice`, and `Quot.sound`. There are no `sorry`, `admit`, custom axioms or prohibited proof escapes. Two genuine failed builds are preserved with diagnosis and repair, and neither altered a mathematical statement [1].

The result brings the reconstruction to a narrow threshold. Levels A and B—pointwise and locally continuous internal representability—were already closed. Task 29 proves that Level C, simultaneous trivialization of all triple defects, is equivalent to Level D, existence of a fully compatible continuous internal transition system. What Task 29 deliberately does *not* prove is that Level C holds for the inherited arbitrary topological family. Nor does it construct the global total space that would result if it did. This distinction is essential: for an arbitrary metric-oriented rank-three family, universal existence of such a lift should not be inferred merely from local solvability.

Part XXX will therefore proceed conditionally. Taking a `CompatibleContinuousInternalTransitions` object as an explicit hypothesis, it should test whether the local products can be glued into a global topological total space, whether the internal group acts freely and fibrewise transitively, and whether the resulting object projects continuously and equivariantly onto the ordinary frame total space. That construction can be settled independently of the still-open existence question. If successful, the project will have reconstructed the complete global object conditional on exactly one remaining compatibility predicate; the later question will then be whether the inherited system satisfies that predicate and how the internally reconstructed criterion compares with the established global obstruction theory.

Keywords: defect-free descent; continuous transition functions; discrete central kernel; $\mathbb{Z}_2$ ambiguity; refinement; simultaneous trivialization; compatible internal transitions; formal verification; spin frontier; Conservation of Difficulty.

Contents

1 The corridor narrows: from a defect to a solvability criterion

The long reconstruction that began with the split-complex Lorentz structure has repeatedly converted broad structural questions into narrower dependency questions. Early stages asked which algebraic carrier was sufficient, which central structure survived closure, how visible rotations were internally represented, and whether globally coherent representatives existed. Later stages separated the visible quotient from its two-valued internal realization, identified the quaternionic core and the visible rotation group, reconstructed the local covering data, built the ordinary oriented-frame torsor and its lifted one-fibre partner, and finally allowed these fibres to vary over a topological base [6, 7].

By Part XXVIII, the ordinary and internal sides had both become mature enough that the remaining incompatibility could be written as an explicit function rather than described qualitatively. On a refined triple overlap, continuous internal representatives satisfy

$$u_{jk}u_{ij} = \delta_{ijk}u_{ik}, \qquad \delta_{ijk} \in K, \tag{1}$$

with $K = \{\pm 1\}$ in the certified instance. The defect is continuous, hence locally constant, and it changes under representative replacement according to

$$\delta'_{ijk} = (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk}. \tag{2}$$

The immediate temptation is to call δ the whole global obstruction and move directly to solving equation (2). Task 29 deliberately refuses that shortcut.

The reason is subtle. The representatives u_{ij} do not initially live on the full overlap $U_i \cap U_j$; they live on an auxiliary refinement $V_{ij\alpha}$. Two different patches for the same ordered pair (i, j) can therefore carry two different continuous representatives of the same ordinary transition. Even if every triple defect vanished, it was not yet formally known that those same-pair representatives must agree. A second descent datum could have survived.

Methodological principle 1 (Do not declare the obstruction complete before descent is hardened)**.** A triple-overlap defect measures failure of composition among three transition labels. A same-pair refinement discrepancy measures failure of two local representatives of one fixed transition to agree. They live in the same kernel but arise from different comparison problems. They may be identified only after a theorem proves the dependence.

Part XXIX therefore begins not by solving the global equation, but by testing whether the defect system is logically complete. The result is unexpectedly economical: the apparent second datum is forced by the first once repeated indices are retained. The corridor to the global problem becomes correspondingly narrower.

2 Formal source, scope, and provenance

The supplied Task-29 archive has SHA-256

`d8d3e62fb1a0eb6237ab499c469db338afded146c4e1defd81db260ee703486c`.

The pinned environment is Lean 4.28.0 with Mathlib commit

`8f9d9cff6bd728b17a24e163c9402775d9e6a365`.

The complete Task-28 source and audit are byte-for-byte identical to the preceding Task-28 archive. Task 29 adds only new source files and documentation.

The new source chain is

`Inherited` → `DefectFreeHardening` → `PairwiseDescent` → `KernelAdjustment` →
`SimultaneousTrivialisation` → `ChoiceInvariance` → `RefinementBehavior` →
`CombinedCompatibility` → `CompatibleTransitions` → `Task29`,

with `Counterexample` branching from `PairwiseDescent` and re-imported by the endpoint.

Check	Result	Detail
Focused Task-29 build	PASS	`lakebuildRequestProject.NullSectorTask29.Task29`, 8257 jobs
Whole-project build	PASS	8328 jobs, 0 errors
Task-29 warnings	0	114 project-wide warnings are inherited
New source	11 Lean files	2217 Lean lines
Axiom audit	59 declarations	each reports only `propext`, `Classical.choice`, `Quot.sound`
`sorry`/`admit`	0/0	no proof holes
Prohibited escapes	0	no custom `axiom`, `implemented_by`, `native_decide`, `bv_decide`, `unsafe`, or `partial`
Genuine failed builds	2	both proof-engineering/elaboration failures; no theorem changed

Proof boundary 1 (Verification boundary)**.** *The supplied source tree, Task-29 audit, endpoint statements, relevant proof bodies, dependency structure and proof-escape scan were inspected directly. The successful Lean builds and* `#print axioms` *outputs are reported from the supplied Aristotle artifact; the paper-generation environment did not independently rebuild the 8,000-plus-job Lean project.*

Task 29 does not construct a global internal total space, does not assign a topology to such a space, and does not construct a global internal-group action over the base. It imports no characteristic class, no cohomology theory, no principal-bundle object, no smooth structure, no connection and no physical spin datum. Those absences are part of the formal scope.

3 Inherited data: local representatives and the triple defect

Let B be the topological base with ordinary transition system

$$g_{ij} : U_i \cap U_j \to G_{\mathrm{vis}},$$

satisfying

$$g_{ii} = 1, \qquad g_{ji} = g_{ij}^{-1}, \qquad g_{ik} = g_{jk} g_{ij}. \tag{3}$$

Let

$$p : \mathcal{L} \longrightarrow G_{\mathrm{vis}}$$

be the certified continuous surjective group homomorphism with discrete central kernel K; in the project instance, $K = \{\pm 1\}$.

Task 28 produced, on an open refinement of each pair overlap, continuous maps

$$u_{ij}^{\alpha} : V_{ij\alpha} \longrightarrow \mathcal{L}, \qquad p(u_{ij}^{\alpha}) = g_{ij}|_{V_{ij\alpha}}.$$

For any three compatible refinement patches, it defined

$$\delta_{ijk}^{\alpha\beta\gamma} = (u_{jk}^{\beta} u_{ij}^{\alpha})(u_{ik}^{\gamma})^{-1}. \tag{4}$$

The ordinary law equation (3) implies that δ is kernel-valued. Continuity and discreteness of the kernel imply local constancy. Under a kernel-valued replacement

$$u'_{ij} = \varepsilon_{ij} u_{ij}, \tag{5}$$

Task 28 proved the exact formula

$$\delta'_{ijk} = (\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk}. \tag{6}$$

It also proved the associativity-induced fourfold relation

$$\delta_{ijk}\delta_{ikl} = \delta_{jkl}\delta_{ijl}. \tag{7}$$

The inherited predicate `IsCompatibleInternalTransitionSystem` requires $\delta = 1$ for all triples and all relevant refinement patches. Task 29 renames this condition `TripleDefectFree` only for clarity; the mathematical content is inherited unchanged.

The key point is the quantifier: *all triples* means that repeated indices such as (i, i, i) and (i, i, j) are included. Task 29 proves that these degenerate-looking instances contain precisely the descent information that was missing from the Part XXVIII interpretation.

4 Hard target A: repeated-index rigidity

4.1 The completely repeated defect gives the identity

For a group-valued map u, the completely repeated algebraic word is

$$(uu)u^{-1} = u.$$

Task 29 isolates this elementary fact as `tripleDefect_repeated`. When the repeated defect is required to equal 1, the representative itself is forced to equal 1.

Theorem 1 (Identity normalization from defect-freeness). *Let C be a refined internal candidate with all triple defects equal to 1. Then every stored representative of the identity transition satisfies*

$$u_{ii}^{\alpha} = 1$$

on its refinement domain.

The theorem is not an added normalization convention. It is a consequence of the same defect equation used for ordinary triples.

4.2 Repeated triples also force inverses

Once theorem 1 is available, the repeated-index triple (i, j, i) gives

$$u_{ji}u_{ij} = u_{ii} = 1.$$

Hence:

Theorem 2 (Inverse normalization). *Under full triple-defect freeness,*

$$u_{ji} = u_{ij}^{-1}$$

whenever the relevant refinement patches meet at the same base point.

For arbitrary (i, j, k) one has immediately:

Theorem 3 (Exact internal triple law). *Full triple-defect freeness is equivalent, on each common refined triple-overlap domain, to*

$$u_{jk}u_{ij} = u_{ik}.$$

Thus identity, inverse and composition are not three independent normalization requirements. They are encoded in one all-indices defect predicate.

Methodological principle 2 (Repeated indices carry structural content). In a transition law quantified over all chart labels, repeated-index instances encode the unit and inverse laws. Removing them changes the mathematics. A "generic distinct-index" restriction is not an equivalent simplification.

5 The apparent second defect: same-pair descent

5.1 Definition and local structure

Take two refinement patches $V_{ij\alpha}$ and $V_{ij\beta}$ carrying continuous representatives of the same ordinary transition. Task 29 defines

$$\kappa_{ij}^{\alpha\beta} := u_{ij}^{\beta}(u_{ij}^{\alpha})^{-1} \tag{8}$$

on their intersection. This is exactly the inherited relative-factor construction applied to two representatives of the same ordinary map.

The resulting theorems parallel the Task-28 ambiguity analysis:

(i) $\kappa_{ij}^{\alpha\beta} \in K$;

(ii) κ is continuous;

(iii) κ is locally constant because K is discrete;

(iv) $\kappa_{ij}^{\alpha\alpha} = 1$;

(v) $\kappa_{ij}^{\beta\alpha} = (\kappa_{ij}^{\alpha\beta})^{-1}$;

(vi) on triple intersections of refinement patches,

$$\kappa_{ij}^{\alpha\gamma} = \kappa_{ij}^{\beta\gamma}\kappa_{ij}^{\alpha\beta}.$$

For the certified projection, each value lies in $\{\pm 1\}$.

At this stage κ could still have been a second independent descent obstruction: δ compares different transitions under composition, whereas κ compares different local realizations of one fixed transition.

5.2 The decisive theorem: δ controls κ

Theorem 4 (Full triple-defect freeness forces pairwise refinement coherence)**.** *If all Task-28 triple defects vanish, including repeated-index instances, then*

$$\kappa_{ij}^{\alpha\beta} = 1$$

on every intersection $V_{ij\alpha} \cap V_{ij\beta}$. Equivalently,

$$u_{ij}^{\alpha} = u_{ij}^{\beta}$$

wherever both are defined.

The proof is short but conceptually decisive. Use the triple (i, i, j). Its defect contains

$$u_{ij}^{\beta}u_{ii}^{\rho}(u_{ij}^{\alpha})^{-1}.$$

By theorem 1, $u_{ii}^{\rho} = 1$. Defect-freeness then gives precisely

$$u_{ij}^{\beta}(u_{ij}^{\alpha})^{-1} = 1.$$

Hence $\kappa = 1$.

The result collapses the feared two-defect problem to a single all-indices defect problem.

5.3 Why this is not tautological: the preserved counterexample

Task 29 permanently preserves a minimal counterexample to the weaker statement in which $\delta = 1$ is required only for pairwise distinct indices. The construction uses:

- a one-point base;
- one chart index;
- a trivial ordinary group;
- a two-element discrete central internal group;
- two refinement patches carrying the two distinct constant representatives 1 and $z \neq 1$.

There are no triples of pairwise distinct indices, so the restricted defect condition is vacuously true. Pairwise refinement coherence nevertheless fails.

Theorem 5 (Distinct-index defect-freeness is insufficient)**.** *The restriction of triple-defect freeness to pairwise distinct chart indices does not imply same-pair descent.*

The counterexample proves that theorem 4 depends genuinely on the repeated-index sector. This is an important instance of Conservation of Difficulty: a seemingly degenerate part of the index domain carries exactly the algebraic information that would otherwise have to be supplied as an independent normalization axiom.

6 From refined representatives back to the original overlaps

Once representatives of one fixed pair agree on intersections of refinement patches, ordinary topological gluing applies.

Theorem 6 (Conditional pairwise gluing)**.** *Assume the stored representatives u_{ij}^{α} agree on every mutual intersection of their refinement patches. Then there exists a unique map*

$$v_{ij} : U_i \cap U_j \longrightarrow \mathcal{L}$$

whose restriction to every $V_{ij\alpha}$ is u_{ij}^{α}. The glued map is continuous and satisfies

$$p(v_{ij}) = g_{ij}.$$

The construction uses a local patch containing each point and classical choice only to define the raw value; pairwise coherence proves independence of the chosen patch. Continuity is local: near any point, the glued function agrees with one stored continuous representative.

Combining theorems 4 and 6 yields a stronger consequence than Part XXVIII could state:

Corollary 1 (Defect-free refined data descend to whole overlaps)**.** *Every fully triple-defect-free refined internal system determines one continuous internal representative on each original overlap $U_i \cap U_j$.*

Moreover, the repeated-index theorems survive the gluing. The resulting whole-overlap maps satisfy

$$v_{ii} = 1, \qquad v_{ji} = v_{ij}^{-1}, \qquad v_{jk} v_{ij} = v_{ik}. \tag{9}$$

This is the first point at which the project has a mathematically complete internal transition-system interface on the original atlas—conditionally on defect-freeness.

7 Hard target B: one kernel adjustment and one equation system

7.1 Admissible adjustments

Task 29 packages a continuous kernel-valued modification on every refined pair domain as a `KernelAdjustment`. If ε is such an adjustment, the representatives change by

$$u^{\alpha}_{ij} \longmapsto (u^{\varepsilon})^{\alpha}_{ij} := \varepsilon^{\alpha}_{ij} u^{\alpha}_{ij}.$$

The ordinary projection is unchanged because ε lies in the kernel. Identity, composition and inverse operations on adjustments are formalized, and applying an adjustment followed by its inverse returns the original candidate.

7.2 The exact simultaneous equation

Using the Task-28 transformation law, Task 29 proves an equivalence rather than a mere necessary condition.

Theorem 7 (Simultaneous trivialization equation)**.** *For a fixed admissible kernel adjustment ε, the adjusted candidate is triple-defect-free if and only if, on every refined triple overlap,*

$$(\varepsilon_{jk}\varepsilon_{ij}\varepsilon_{ik}^{-1})\delta_{ijk} = 1. \tag{10}$$

Equivalently,

$$\varepsilon_{ik} = \varepsilon_{jk}\varepsilon_{ij}\delta_{ijk}. \tag{11}$$

Task 29 defines `CanTrivialiseC` to mean existence of one admissible adjustment satisfying this condition. It then proves that this predicate is exactly equivalent to the inherited Task-28 predicate `CanTrivialiseSimultaneously`; the change is an interface hardening, not a new mathematical assumption.

7.3 No second adjustment is required

A priori, one adjustment might have solved the triple equations while another adjustment was required to make same-pair representatives descend. That would have left a coupled two-equation problem. Theorem 4 removes this possibility.

Theorem 8 (One adjustment solves every discovered compatibility condition)**.** *There exists one admissible kernel adjustment making the system compatibility-ready if and only if there exists one adjustment for which*

(i) all triple defects vanish, and

(ii) all same-pair refinement discrepancies vanish.

The second property follows automatically from the first.

Hence the project does not accumulate a hierarchy of independent signs. The all-indices triple-defect system remains the unique compatibility equation discovered at this layer.

Methodological principle 3 (Conservation of Difficulty is not proliferation of defects)**.** A deeper decomposition may reveal a new datum, but that datum must be tested for logical independence. Task 29 reconstructs κ and then proves it dependent on δ. The correct dependency graph therefore contracts again rather than growing monotonically.

8 Hard target C: choice and refinement do not change the verdict

A global compatibility predicate is meaningful only if it is independent of arbitrary local presentation choices. Task 29 proves this at three levels.

8.1 Initial representative choice

Suppose the same refined domains carry another family of continuous internal representatives of the same ordinary transitions. The two families differ by continuous kernel-valued functions. Task 29 proves:

Theorem 9 (Representative-choice invariance)**.** *Simultaneous trivializability is unchanged by replacing the initial local internal representatives with any other admissible representatives on the same refined domains.*

The proof composes the initial change with a prospective trivializing adjustment. Thus the solvability verdict is not a property of one arbitrary local lift choice.

Task 29 also defines the minimal relation `DefectGaugeRelated`: two adjustment choices are related if one is obtained from the other by a further admissible kernel adjustment. Reflexivity, symmetry and transitivity are proved, and trivializability is constant on this relation. No quotient and no external name are introduced.

8.2 Refinement invariance

Restriction of a solvable candidate to a finer open refinement is immediately solvable by restriction of the same kernel adjustment. The converse is less obvious: solvability after refining the cover need not generally imply solvability before refinement unless one has a descent theorem.

Task 29 proves the converse indirectly but rigorously. Once the main whole-overlap transition theorem of the next section is available, both the original candidate and every refinement are solvable exactly when the same global-on-overlaps transition object exists. Therefore:

Theorem 10 (Refinement invariance)**.** *For every candidate C and every admissible refinement R,*

$$\mathrm{CanTrivialise}(C|_R) \iff \mathrm{CanTrivialise}(C).$$

Consequently, "trivializable after refinement" is not a weaker notion in this formal setting.

This theorem does not claim a general cofinality principle for arbitrary cover-based obstruction theories. It is a consequence of the specific descent theorem proved here: full defect-freeness forces same-pair agreement and hence gluing to the original overlaps.

8.3 Candidate independence

The strongest statement is then immediate.

Theorem 11 (Candidate independence)**.** *Any two refined internal candidates over the same ordinary transition system and the same internal projection are simultaneously trivializable together:*

$$\mathrm{CanTrivialise}(C) \iff \mathrm{CanTrivialise}(C').$$

Thus the unresolved existence question is now a property of the underlying ordinary transition system relative to the internal projection, not a property of auxiliary representatives or refinements.

9 The main equivalence: solvability exactly means compatible transitions exist

Task 29 packages the desired whole-overlap object as `CompatibleContinuousInternalTransitions`. It contains one map

$$v_{ij} : U_i \cap U_j \to \mathcal{L}$$

for each ordered pair of ordinary charts, with the following fields:

$$v_{ij} \text{ is continuous,} \tag{12}$$
$$p(v_{ij}) = g_{ij}, \tag{13}$$
$$v_{ii} = 1, \tag{14}$$
$$v_{ji} = v_{ij}^{-1}, \tag{15}$$
$$v_{jk}v_{ij} = v_{ik}. \tag{16}$$

Nothing resembling a total space, bundle projection or global group action is included in this structure.

Theorem 12 (Principal Task-29 equivalence)**.** *For every refined internal candidate C over the fixed ordinary transition system,*

$$\mathrm{CanTrivialise}(C) \iff \mathrm{Nonempty}\big(\mathrm{CompatibleContinuousInternalTransitions}\big).$$

Equivalently, the inherited Task-28 predicate `CanTrivialiseSimultaneously` *holds exactly when a normalized continuous internal transition system exists on the original overlaps.*

The two directions reveal why the theorem is strong.

Forward direction. A solving adjustment makes the refined candidate defect-free. Repeated-index rigidity then supplies identity and inverse normalization; full defect-freeness forces same-pair coherence; the pairwise gluing theorem produces one continuous representative on each original overlap; and the exact triple law descends to the glued maps.

Backward direction. Assume a compatible whole-overlap system v_{ij} already exists. Compare each stored refined representative u_{ij}^{α} with v_{ij}. Their relative factor is continuous and kernel-valued. Using these factors as the adjustment turns every stored representative into the restriction of v_{ij}, and the exact transition law for v forces every adjusted triple defect to vanish.

The theorem closes the logical gap that remained after Part XXVIII. There is now no ambiguity about what solving the kernel equations buys: it buys exactly the continuous normalized internal transition system that will be needed for a global gluing construction.

Status 1 (The compatibility layer is complete, the existence verdict is not)**.** Task 29 proves an exact equivalence between the explicit kernel-equation problem and existence of compatible internal transitions. It does *not* prove either side for the inherited arbitrary metric-oriented family. The remaining problem is existence, not definition.

10 The level hierarchy after Task 29

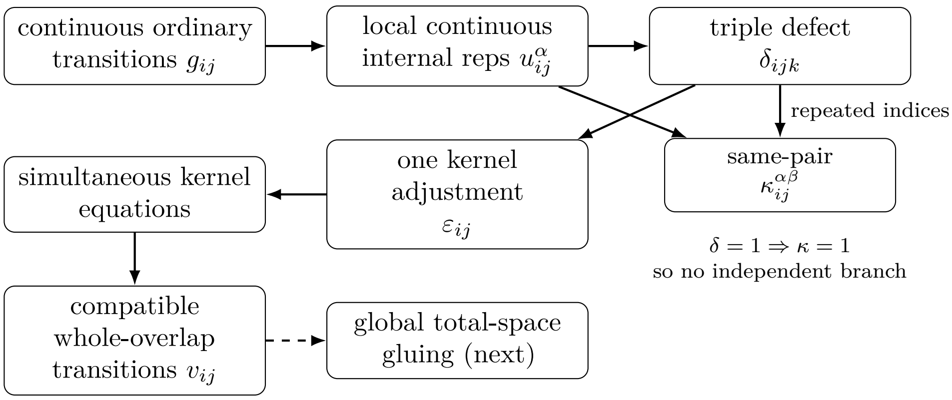
The project can now state the global problem without conflating distinct notions of existence.

Level	Statement	Status
A	For each point and each ordinary transition value, an internal representative exists.	inherited, proved
B	After refinement, every ordinary transition admits local continuous internal representatives.	inherited from XXVIII, proved
C	One continuous kernel adjustment simultaneously removes every triple defect.	characterized, existence open
D	One adjustment satisfies every descent requirement; equivalently, compatible whole-overlap internal transitions exist.	C $\Leftrightarrow$ D proved
E	The compatible transitions are glued into a global topological total space with the expected internal action and projection to the ordinary frame total space.	not constructed

The collapse $C \Leftrightarrow D$ is the central achievement of Task 29. It means that there is no hidden local-descent layer between the explicit kernel equations and the transition data required for a global construction.

11 Dependency contraction and Conservation of Difficulty

The recent part of the project can now be summarized by a dependency graph in which some branches genuinely disappear after hardening:



The new Conservation-of-Difficulty statement is therefore unusually sharp:

Methodological principle 4 (Difficulty has contracted to existence)**.** The project no longer lacks the correct local representatives, the correct defect, the correct descent law, or the correct transition-system target. All of those have been reconstructed and related by equivalence theorems. The remaining compatibility difficulty is whether the explicit kernel equations admit a solution for the inherited global data.

This is a small node in the dependency graph but not a minor mathematical question. Local solvability has repeatedly survived, while global coherent choice is precisely where the nontriviality has accumulated.

12 Late comparison with established mathematics

Only after the intrinsic Task-29 structure is fixed is it useful to compare it with the standard language of spin geometry and bundle theory [4, 5, 8, 3].

For an oriented rank-three metric vector bundle in the conventional setting, the oriented orthonormal frame bundle has structural group $SO(3)$. A spin structure consists, in one standard formulation, of a principal Spin(3) bundle together with a two-to-one equivariant map to the oriented frame bundle covering the identity on the base and compatible with the double covering $\mathrm{Spin}(3) \to SO(3)$ [4]. On a sufficiently fine local trivializing cover, such a structure can be described by transition maps into Spin(3) projecting to the ordinary $SO(3)$ transitions and satisfying the exact overlap law.

The present project has independently reconstructed nearly every ingredient appearing in that local description:

- the visible group is formally group-equivalent to $SO(3)$;
- the internal core is formally group-equivalent to the unit quaternions and to $SU(2)$;
- the internal projection agrees with the quaternionic rotation map and has exact kernel $\{\pm 1\}$;
- ordinary oriented frames form a free transitive visible-group torsor;
- lifted frames form a free transitive internal-group torsor;
- a varying topological frame family with continuous ordinary transition maps has been reconstructed;
- those transitions are locally continuously representable internally;
- their ambiguity and triple-overlap defect are exactly kernel-valued;
- Task 29 now proves that solving the kernel equations is equivalent to existence of exact compatible internal transition maps on the original overlaps.

The resemblance to the standard transition-function formulation is therefore no longer merely qualitative. Nevertheless two formal boundaries remain decisive.

First, the project has not proved that the inherited arbitrary metric-oriented rank-three family admits compatible internal transitions. In established mathematics one does not expect universal existence for arbitrary oriented rank-three bundles over arbitrary bases. The existence condition is conventionally encoded by the vanishing of the second Stiefel–Whitney class. The formulas reconstructed in Parts XXVIII–XXIX have the familiar shape of a $\mathbb{Z}_2$-valued degree-two transition defect modified by degree-one sign choices, but the Lean project has not introduced Čech cohomology, has not constructed w_2, and has not proved that its internally derived defect-gauge relation represents the classical w_2 obstruction [5, 2].

Second, even if compatible internal transitions are supplied, Task 29 has not yet glued them into a global total space. Therefore it has not yet produced an object satisfying the global bundle-level definition of a spin structure.

Proof boundary 2 (What may and may not be claimed after Part XXIX)**.** *It is justified to say that the project has reconstructed the exact local continuous transition criterion required for a spin-type global lift and has reduced its solvability to one explicit kernel equation system. It is not yet justified to say that the arbitrary inherited family possesses a spin structure, nor that the reconstructed defect has formally been identified with w_2.*

The distinction also clarifies what a possible "completion" of the present research arc would mean. Completion need not be a theorem that every arbitrary input family is liftable. A mathematically complete reconstruction could instead consist of: an intrinsic existence criterion, a conditional global construction when the criterion holds, and a late theorem identifying that criterion with the established definition and obstruction theory.

13 Open obligations after Part XXIX

The formal audit leaves four principal obligations.

13.1 The actual Level-C existence verdict remains open

For the inherited ordinary transition system from an arbitrary topological metric-oriented rank-three family, Task 29 proves

$$\text{CanTrivialiseSimultaneously} \iff \text{Nonempty}(\text{CompatibleContinuousInternalTransitions}),$$

but proves neither side unconditionally. This is the single remaining compatibility-existence question.

13.2 No global obstruction invariant has been constructed

The project uses the explicit defect-change relation and proves representative and refinement invariance of the solvability verdict. It does not quotient the defect systems, assign a class to them, or compare such a class formally with any standard invariant.

13.3 No classification of the solution set is attempted

No connectedness, paracompactness, separation or local compactness assumption enters the Task-29 theorem. Consequently the task does not classify how many compatible transition systems may exist when one exists, nor does it identify a torsor of global choices.

13.4 The global total-space layer is untouched

The local transition data required for gluing are now frozen, but no equivalence relation on a disjoint union of local products, no quotient topology, no global projection, and no global internal action have yet been constructed.

14 Part XXX: the conditional global construction

The next task should deliberately separate construction from existence. It should take as an explicit hypothesis

$$T : \text{CompatibleContinuousInternalTransitions}(p, S)$$

and ask only whether these data are sufficient to build the expected global topological object. In particular, Part XXX should not attempt to solve the Task-29 kernel equations for the inherited family.

The focused conditional program is:

1. **Local product carrier.** Form the disjoint union of local products $U_i \times \mathcal{L}$ over the ordinary atlas. Derive the correct identification relation on pair overlaps directly from the multiplication convention in $T.v_{ij}$; do not assume left-versus-right multiplication.

2. **Gluing relation.** Prove reflexivity, symmetry and transitivity from

$$v_{ii} = 1, \qquad v_{ji} = v_{ij}^{-1}, \qquad v_{jk}v_{ij} = v_{ik}.$$

   Construct the quotient carrier only after the relation is proved.

3. **Topology and local triviality.** Give the quotient/atlas-generated total space a topology for which the canonical local product maps are homeomorphisms. Prove the projection to B continuous and identify every fibre with the internal group at the strongest justified level.
4. **Global internal action.** Determine the multiplication side that descends to a well-defined continuous free fibrewise transitive action of $\mathcal{L}$ on the glued total space.
5. **Map to the ordinary frame total space.** Use the already certified projection $p : \mathcal{L} \to G_{\mathrm{vis}}$ and the ordinary frame trivializations to construct a global map from the new total space to the ordinary oriented-frame total space. Prove well-definedness, continuity and equivariance rather than assuming them from the local formulas.
6. **Two-sheeted fibre relation.** Specialize the kernel to the certified $\{\pm 1\}$ and prove the strongest correct global statement about fibres of the map to the ordinary frame space. Do not import a named structure merely because the resulting diagram looks familiar.
7. **Late comparison only.** If the construction closes, compare the resulting global diagram with the standard definition of a spin structure. Keep the comparison conditional on the input T and separate it from the unresolved theorem that such a T exists for the inherited family.

The dependency split is deliberate:

$$\boxed{\text{Task XXIX: existence criterion}}$$

versus

$$\boxed{\text{Task XXX: conditional object construction}}$$

Proving the second direction first has methodological value. It establishes exactly what the solution of the kernel equations would buy and prevents the later existence analysis from targeting an underspecified object.

Methodological principle 5 (Conditional construction before global existence classification)**.** Once a precise compatibility interface has been reconstructed, the global object it would generate can be tested independently of whether the interface is inhabited for a particular input. This separates a geometric gluing theorem from an obstruction/existence theorem and keeps the dependency graph acyclic and explicit.

If Part XXX succeeds, the reconstruction will stand immediately before a possible closure of the present arc: the global object will be formally available whenever the Task-29 criterion holds, while one remaining question will determine when it holds for the inherited family. At that point a final comparison can ask whether the intrinsically reconstructed conditional object and its existence criterion coincide exactly with the established global spin structure and its classical obstruction.

15 Conclusion

Part XXIX begins with a genuine risk that the Task-28 triple defect is incomplete. The local representatives live on refined overlap patches, so a second same-pair descent defect could have survived independently. The task reconstructs that candidate datum κ rather than assuming it away. The result is a strong contraction: full triple-defect freeness, because it includes repeated-index triples, forces $u_{ii} = 1$, the inverse law, exact triple multiplication and $\kappa = 1$. A minimal counterexample proves that this contraction would fail if repeated indices were discarded.

The refinement patches can therefore be glued back to the original overlaps once the triple defects vanish. Kernel adjustments are then shown to solve every discovered compatibility requirement through one explicit equation system. Solvability is independent of the initial representatives, invariant under refinement in both directions, and ultimately independent of the chosen refined candidate. The principal theorem identifies that solvability exactly with existence of normalized continuous internal transitions on the original overlaps.

This is a near-terminal result for the local and transition-theoretic side of the project. No hidden descent layer remains between the defect equations and the global gluing input. Yet the last compatibility question is also the one that cannot be dismissed by local arguments: whether the equations possess a solution for the inherited global data. The project is therefore simultaneously close to a formal global object and still separated from it by the precise global existence issue that the preceding reconstruction was designed to expose.

The next step will not blur those two facts. Part XXX will assume compatible internal transitions and test the global gluing construction conditionally. Only after the resulting total-space diagram is formally closed should the project return to the unresolved existence predicate and, finally, compare the internally reconstructed criterion with the established global mathematics.

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